

Project 18

Local versus Global Joint-Limit Avoidance on iiwa 7

Format: Group of two students; simulation-based project.

Prerequisites: Redundant inverse kinematics, null-space projection, joint-limit avoidance, and feasibility-map planning.

Motivation

Null-space motion can improve a local joint-centering objective while missing a feasible route elsewhere in the redundant IK set. Comparing that local strategy with a global feasibility chart helps distinguish algorithmic limitations from actual restrictions imposed by the robot and the task.

Task

Overall objectives. Compare local null-space joint centering with planning through a classified iiwa 7 IK chart, and explain success or failure through feasible self-motion intervals and their connections along the path.

- **Establish one common problem.** Validate the iiwa model, limits, and complete pose path. Use one short path segment that approaches a joint limit and two admissible initial configurations, ensuring both methods are compared from the same initial configuration in each trial.
- **Implement local joint centering.** Use the geometric Jacobian with a documented scaling and a projected gradient for $h(q) = \frac{1}{2} \sum_i ((q_i - q_{i,c})/r_i)^2$. Integrate small steps, correct the full-pose error, and recheck limits. Explain why quadratic centering is a preference, not a hard limit barrier.
- **Build the global comparison chart.** Enumerate the supplied fixed- q_3 IK families on path-sample versus redundant-angle axes. Preserve branch labels, plot feasible intervals and failure reasons, and distinguish full-Jacobian rank loss from reduced-chart degeneracy.
- **Plan under the same constraints.** Select a chart corridor from the same starting configuration, using joint centering and joint travel as documented costs. Validate intermediate transitions and apply the same pose, limit, and regularity tolerances used for the local method.
- **Diagnose the difference.** Overlay both trajectories on the classification charts. Compare completion, minimum limit margin, travel, and pose error for the two initial states. Repeat the local method with one smaller integration step to distinguish discretization error from failure to discover another feasible corridor.

Specifics and scope

Use one path segment, two initial configurations, and two local step sizes. Start with about 120 path samples and 120 redundancy angles and refine decisive intervals. Reuse the supplied analytical IK and construct the chart by enumerating and validating candidates on the path–redundancy grid; a complete global optimizer is unnecessary. Equal results are acceptable when the explanation is supported by the feasibility maps.

For every chart, keep pose validity, joint limits, full-Jacobian regularity, and reduced-chart regularity as separate diagnostics. The rectangular 6×7 Jacobian has no determinant. A feasible pixel is not a validated edge: re-solve intermediate prescribed poses and check joint continuity within limits. If collision checking is unavailable, state that feasibility is kinematic only; a clearance proxy is not a full collision test.

Expected outcome

Understand the difference between instantaneous null-space freedom and global feasible connectivity. Deliver classified redundancy charts with trajectory overlays, matched comparisons, and an explanation of whether observed limitations arise from the strategy, numerical steps, or constraints.

Submit reproducible code, model parameters and test data, the required plots, a concise 5–6-page report, and a short simulation demonstration. Explain both successful cases and failures; conclusions must follow the numerical and geometric evidence rather than an expected outcome.

Resources

The KUKA LBR iiwa 7 robot description (URDF) and visualization meshes (STL) will be provided to students, together with a mathematical continuation model with unrestricted joints and an analytical fixed- q_3 IK model. Reuse the supplied IK rather than deriving a general 7R solver. Apply the physical joint limits from the robot description separately when using the unrestricted model. Check D–H parameters, joint axes, and the fixed tool transform against the URDF, and validate IK outputs by independent forward kinematics.

A sampled rectangular full-pose path and a reference fixed- q_3 IK table will also be provided. The pose data contain position coordinates and quaternion columns $\mathbf{qw}, \mathbf{qx}, \mathbf{qy}, \mathbf{qz}$; normalize the quaternions and make their signs consistent before interpolation. The reference IK table checks solutions at $q_3 = 30^\circ$ only; it does not replace evaluation at other angles. Use its configurations to check pose residuals and branch labels, applying physical joint limits separately.

Implement the null-space controller and feasibility-chart comparison using the supplied IK and model. Choose a short segment or documented placement of the supplied path; a suitable joint-limit example can be supplied to keep scenario construction bounded. Visualize both recovered trajectories with the supplied meshes and use identical validation tolerances.